

HeyAnna

Technical Product Whitepaper

The Professional Intelligence Terminal for Prediction Markets

Version 1.1

Abstract

This document describes the technical architecture and product design of HeyAnna, an execution-intelligence terminal for prediction markets. It explains how the system aggregates market data, computes derived signals, monitors external information sources, and delivers actionable intelligence to traders.

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1 Introduction

Prediction markets have matured into a globally distributed financial system in which contracts represent the probabilistic outcomes of real-world events.

Platforms such as Polymarket, Kalshi, and Manifold have introduced liquid markets around politics, macroeconomics, geopolitics, sports, and cryptocurrency. However, these markets remain **operationally fragmented**. Each venue exposes independent APIs, pricing structures, liquidity pools, and settlement mechanics.

As a result, professional participants operate across multiple venues simultaneously while lacking:

- unified market visibility
- cross-platform exposure tracking
- automated arbitrage detection
- event-driven execution intelligence

HeyAnna is an **execution-intelligence terminal** designed to unify prediction-market infrastructure into a single operational layer.

The system aggregates market data, computes derived signals, monitors external information sources, and surfaces actionable intelligence to traders through a real-time analytics interface and automated alert infrastructure.

2 Product Overview

HeyAnna functions as an **intelligence and monitoring layer** on top of multiple prediction market venues.

The product performs four primary functions:

1. **Market Aggregation:** Continuous ingestion of price, orderbook, and volume data from prediction market platforms.
2. **Signal Computation:** Detection of arbitrage, liquidity shifts, correlation anomalies, and pricing inefficiencies.
3. **Event Monitoring:** Real-time analysis of global information sources to detect events that impact market probabilities.
4. **Execution Intelligence Delivery:** Delivery of actionable alerts and analytics to users through terminal dashboards and automated voice notifications.

Rather than acting as a trading venue itself, HeyAnna operates as a **market intelligence and execution coordination system**.

3 System Architecture

HeyAnna is built as a **distributed data processing system optimised for low-latency financial intelligence**.

The architecture consists of five primary layers.

3.1 Data Ingestion Layer

The ingestion layer is responsible for collecting raw data from prediction market platforms and external information feeds.

Sources include:

- platform APIs (REST + WebSocket)
- scraping infrastructure for non-API markets
- financial and geopolitical news feeds
- social signal streams
- blockchain event data

Incoming data streams include:

- market prices
- order book states
- trade events
- contract metadata
- liquidity depth
- settlement outcomes

Data ingestion operates through **persistent WebSocket streams** where available to minimise polling latency.

All incoming messages are normalised into a **common market schema** before entering the analytics pipeline.

Key properties:

- horizontal ingestion workers
- platform-specific adapters
- message validation
- timestamp synchronization

This layer is designed to handle **tens of thousands of updates per second**.

3.2 Market Normalisation Engine

Prediction market platforms represent contracts differently. For example:

Platform	Contract Model
Polymarket	YES/NO tokenized outcomes
Kalshi	Binary contracts with CFTC regulation
Manifold	Community-driven probability markets

Table 1: Contract representation across platforms

HeyAnna converts all market representations into a **unified probabilistic contract structure**.

Normalised fields include:

- outcome probability
- implied odds
- market liquidity
- contract expiry
- resolution mechanism

This normalisation allows cross-platform analytics to operate on a **single market abstraction layer**.

3.3 Analytics Engine

The analytics engine continuously processes normalised data streams to produce real-time signals.

Primary analytics modules include:

3.3.1 Arbitrage Detection

Cross-platform price differences are evaluated continuously.

Example:

Platform A: Event YES = 0.68
Platform B: Event YES = 0.74

If transaction costs and liquidity allow profitable execution, the engine flags the opportunity.

Arbitrage detection operates on **sub-second evaluation cycles**.

3.3.2 Liquidity Flow Monitoring

Large order flows often precede price movements.

The system monitors:

- large trades
- order book imbalance
- liquidity withdrawals
- sudden volume spikes

These signals are converted into **flow indicators** visible in the terminal interface.

3.3.3 Correlation Analysis

Prediction markets frequently contain related contracts across platforms.

Example:

- “Will the candidate win the election?”
- “Will the party control the senate?”
- “Will the candidate win a specific state?”

HeyAnna continuously computes correlation matrices to detect mispriced relationships between markets.

3.3.4 Historical Resolution Intelligence

The system maintains a database of resolved markets, including:

- event category
- probability path
- time-to-resolution
- final outcome

This historical dataset allows comparison between current pricing and historical patterns.

3.4 AI Event Intelligence Layer

A major challenge in prediction markets is **event detection**.

Markets move when real-world information changes probabilities.

HeyAnna includes an AI pipeline designed to identify these events in real time.

The system performs four tasks:

1. News ingestion
2. Event classification
3. Position relevance mapping
4. Impact estimation

3.4.1 Event Classification

Incoming news is processed using a fine-tuned language model that categorises events such as:

- geopolitical escalation
- macroeconomic release
- regulatory announcement
- political development
- sports injury update

Each event receives a **confidence score representing potential market relevance**.

3.4.2 Position Mapping

The system maintains a mapping between:

```
event topics -> affected prediction markets
```

Example:

```
Middle East escalation
-> Israel election markets
-> regional stability contracts
-> oil price related prediction markets
```

When an event exceeds the relevance threshold and maps to open positions, the alert pipeline activates.

3.4.3 Impact Estimation

A quantification model evaluates the potential probability shift using historical event-impact data.

Outputs include:

- expected price movement
- exposure size
- urgency level

3.5 Alert and Notification Infrastructure

Once a signal is generated, the system dispatches alerts through multiple channels.

Alert types include:

- terminal push notifications
- Telegram voice messages
- automated phone calls
- email alerts

Critical alerts aim for **sub-10-second end-to-end latency**.

The delivery system includes redundant messaging providers to maintain reliability.

4 Terminal Interface

The HeyAnna terminal provides a consolidated operational interface for prediction market participants.

Core modules include:

4.1 Global Market Feed

Displays real-time pricing across all integrated platforms.

Users can filter markets by:

- category
- event type
- liquidity
- time to resolution

4.2 Arbitrage Monitor

A continuously updating panel listing cross-platform price discrepancies.

Signals include:

- profit margin estimate

- available liquidity
- execution difficulty score

4.3 Portfolio Exposure Dashboard

Users connect their trading accounts to HeyAnna through API keys.

The system aggregates positions across platforms and displays:

- total exposure
- event concentration
- correlated risk

4.4 Event Radar

The AI agent's detected signals appear as a chronological feed showing:

- detected event
- impacted markets
- probability shift estimate
- suggested review urgency

5 Revenue Generation via Execution Intelligence

HeyAnna does not operate as a prediction market.

Revenue is generated through **execution intelligence usage**.

The system produces value through:

- arbitrage discovery
- event-driven trade signals
- liquidity flow detection
- data infrastructure

Execution-driven revenue arises from three mechanisms.

5.1 Data Access

Users pay for access to:

- real-time cross-platform market data
- arbitrage detection signals
- historical market datasets

5.2 Signal Infrastructure

Advanced analytics and AI-generated signals are available through the terminal interface and APIs.

These signals are generated continuously and represent the core product output.

5.3 Execution-Linked Services

Some integrations enable automated or semi-automated execution workflows where HeyAnna coordinates data and signals with trading systems.

In these scenarios, the platform receives **a small fee tied to executed signals or API usage**.

6 Reliability and Scalability

Prediction market opportunities exist on short time horizons. Infrastructure reliability is therefore critical.

The system is designed for:

- high-availability ingestion
- redundant alert pipelines
- distributed computation

Key engineering principles include:

- horizontal scaling
- streaming data pipelines
- event-driven microservices
- version-controlled analytics models

All models and signals are logged to enable reproducibility and auditability.

7 Future Development

Future technical expansion includes:

- Additional prediction market platform integrations
- Advanced probability modelling
- Automated execution routing
- Institutional data APIs
- Predictive signal training on expanding event datasets

The objective is to build a continuously improving intelligence system for probabilistic markets.

8 Conclusion

Prediction markets are emerging as a new financial category where probabilities are directly traded.

However, the ecosystem lacks the infrastructure required for professional participation.

HeyAnna addresses this gap by building a **unified intelligence terminal** that aggregates market data, computes analytics, monitors real-world events, and delivers actionable signals in real time.

The system functions as the operational layer that enables participants to **see information faster, interpret it accurately, and act before markets fully reprice**.